

IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

INTERNSHIP REPORT OF C PROGRAMMING VIRTUAL INTERNSHIP

Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Technology - Computer Science and Engineering

Submitted By

SWASTIK BARMAN

Roll No.: **25SCS1003002361**

Batch: 2026-27

Organization / Platform: **BeeSkilled**

Internship Domain: **C Programming**

Internship Duration: **6-Week Online Internship Program**

CANDIDATE'S DECLARATION

I, **SWASTIK BARMAN**, Roll No. **25SCS1003002361**, hereby declare that this report titled "C Programming Virtual Internship: Core Programming and Practical Development" has been prepared as an academic record of my learning and practical programming activities for the B.Tech CSE programme.

- The report documents C programming concepts, practical exercises and a small project developed as part of the learning experience.
- Technical references used in preparing the report have been acknowledged.

Signature: _____

Name: **SWASTIK BARMAN**

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ACKNOWLEDGEMENT

I express my sincere gratitude to everyone who supported me during the internship learning program. I am thankful for their guidance, encouragement and constructive feedback throughout the completion of this work.

I would like to thank BeeSkilled for providing an online learning environment and resources through which I could study C programming, problem-solving techniques and fundamental programming concepts. The practical orientation of the learning activities helped me connect programming theory with implementation.

I also thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their academic support and encouragement. I am grateful to my parents for their continuous support and motivation.

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4. PROJECT DESCRIPTION

4.1 Introduction

This report documents the learning and practical work undertaken during a C programming internship/training program. The internship focused on building a strong foundation in C syntax, procedural programming, problem-solving and structured program development. The learning activities combined conceptual study with coding exercises designed to improve logical thinking and programming ability.

The practical work covered variables and data types, operators, conditional statements, loops, functions, arrays, strings, pointers, structures and file-oriented programming concepts. A small console-based Student Record Management System was selected as an integrated project to demonstrate these concepts.

4.2 Organization Profile

BeeSkilled is used in this report as the online internship/training platform through which the C programming learning activities were undertaken. The platform provides skill-oriented learning and internship activities intended to give students exposure to practical technical concepts.

4.3 Problem Statement

The practical problem addressed was to develop the ability to translate programming requirements into structured C solutions. The internship therefore focused on incremental implementation, testing and debugging, allowing individual concepts to be combined into a complete small application.

4.4 Project Objectives

- To understand C syntax, variables, operators and control structures.
- To develop logical and algorithmic problem-solving skills.
- To understand functions, arrays, strings, pointers and structures.
- To practise modular programming and structured program design.
- To practise compilation, debugging and testing.
- To integrate the concepts into a small console-based C project.

4.5 Scope of the Project

The scope is educational and focuses on core C programming rather than production-scale software engineering. The application demonstrates basic data handling, menu-driven interaction and structured programming. It is a learning prototype and does not include database integration, cloud deployment or large-scale distributed architecture.

4.6 Technologies and Tools Used

- **C:** Primary programming language used for implementation.
- **C Compiler:** Compilation and identification of syntax errors.
- **IDE/Code Editor:** Writing, running and debugging C programs.
- **Standard C Library:** Input/output, string handling and other utility functions.
- **BeeSkilled Learning Platform:** Accessing internship/training material and activities.

4.7 System Architecture

The project follows a simple menu-driven architecture. User input is accepted through the console, validated and processed using the appropriate function, and the result is displayed back to the user.

- **Input:** Accept student details or menu selection.
- **Validation:** Check basic input conditions.
- **Processing:** Execute the selected function and update records.
- **Output:** Display records, search results or calculated statistics.
- **Testing:** Test different inputs and correct errors.

4.8 Methodology

The internship methodology follows a learn-implement-test-debug cycle. Each concept is first understood through learning material, then implemented in a short program. The program is compiled and tested with multiple inputs. Errors are identified, corrected and the solution is reviewed for clarity and modularity.

- **4.8.1 C Fundamentals:** The initial stage covered program structure, header files, the main function, variables, constants, primitive data types, input/output statements and operators. Conditional statements and loops were used to create decision-making and repetitive logic.

- **4.8.2 Functions, Arrays and Strings:** Functions were used to divide programs into reusable logical units. Arrays were introduced for storing collections of values, while strings were used for text processing. Parameters and return values were practised to improve program organization.
- **4.8.3 Pointers and Structures:** Pointers were studied for working with memory addresses and for passing data efficiently to functions. Structures were used to group related data items such as a student's name, roll number and marks into a single record. These concepts supported modular and structured program design.
- **4.8.4 Testing and Debugging:** Programs were tested using normal, boundary and invalid inputs. Compilation errors, runtime issues and logical errors were identified through systematic testing and debugging.

4.9 Expected Outcomes

- A working foundation in C programming and syntax.
- Improved logical reasoning and problem-solving skills.
- Understanding of structured programming concepts.
- Ability to compile, test and debug basic C applications.
- A small console-based project integrating multiple concepts.
- A foundation for further study of Data Structures and Algorithms using C.

4.10 Internship Training Log Book

The schedule is an academic reporting structure adapted from the provided internship report and converted from C++ to C. The certificate confirms completion of a 6-week online internship program.

Week	Activities	Key Learning Outcome
Week 1	Introduction to C; program structure; variables and data types; operators	Frqut atipotirifc lsyntawitobotlpikovaasid bronsdesprogramm
Week 2	Functions; arrays; strings: formatted input/output; modular program design	Structured programming skills and reusable functions.
Week 3	Pointers; addresses; pointer-based function arguments; dynamic memory	Boetterepitsjetetangjingof memory and efficient data handlin
Week 4	Structures; arrays of structures; record-oriented data handling; file concep	Ability to model and manage grouped records in C.
Week 5	Menu-driven -programming; validation; testing; debugging; integration of m	aqalles multiple C concepts to a complete small applicatio
Week 6	Student Record Management System; final testing; review and documentati	egrated project experience and improved problem-solvin

4.11 Project: Student Record Management System

The integrated project is a menu-driven C console application designed to maintain basic student records. It demonstrates structures, functions, conditional statements, loops, arrays and simple data storage. The application is intentionally small so that each component can be understood and tested.

- **Add Student:** Accept student name, roll number and marks.
- **Display Students:** Display stored student information.
- **Search Student:** Search using the roll number.
- **Calculate Average:** Calculate average marks for stored records.

- **Menu / Exit:** Provide navigation and safely terminate the application.

4.11.1 Testing Strategy: The project can be tested using valid student details, repeated records, boundary marks and invalid menu selections. Each function is tested independently before integration into the complete menu-driven application.

4.11.2 Result: The project demonstrates how fundamental C concepts can be combined into a coherent software solution and provides practical exposure to modular programming, structured data modelling, input handling and debugging.

5. BIBLIOGRAPHY/REFERENCES

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Student Details

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